GATEFALL

GAME DESIGN DOCUMENT / MOBILE KICKOFF / ASSIGNMENT 8-9

01 / TITLE + PITCH

Gatefall is a short 2D survival game: dodge falling dangers, collect coins and hearts, and stay alive until the exit gate opens after 60 seconds.

02 / GENRE + PLATFORM

Single-player arcade survival platformer. Android first, landscape only; iOS device profiles are used for layout testing. Target session: 60 seconds plus menu and replay. Built in Unity 6 with C#.

03 / DIMENSION

2D side view with a pixel-art hero and sprite animation. One camera shows the whole arena, so touch controls stay predictable. No rigged 3D character is needed.

04 / CORE LOOP

MOVE & DODGE -> COLLECT -> SURVIVE -> REPLAY.
Move between platforms, avoid rocks and bombs, pick up hearts and coins, and reach the 60-second goal. The gate changes from closed to open; death restarts the run. Shots currently demonstrate firing and cooldown; projectile damage is outside this MVP.

05 / CONTROLS

Bottom-left: LEFT and RIGHT buttons, held to move. Bottom-right: JUMP (one grounded jump) and FIRE (hold for a shot every 0.3 seconds). Top-right: PAUSE. Menu: PLAY and QUIT. Pause: RESUME, RESTART, MENU. Desktop fallback: A/D or arrows, Space, X. All touch targets are at least 120 x 120 reference pixels.

06 / CONTENT

MVP: 1 arena, 2 raised platforms, 1 hero with Idle/Walk/Jump, 3 falling types (rock: -1 HP; bomb: -2 HP and a delayed explosion; heart: +1 HP), 3 coin pickups and 1 exit gate. Start with 3 HP and 2 seconds of invulnerability after damage. A single patrolling enemy is planned for the later enemy-AI assignment, not required in the kickoff build.

SCOPE & DELIVERY

A SMALL GAME THAT CAN BE FINISHED AND EXPLAINED

07 / ART + SOUND

Use the instructor's OR1 animation kit for the hero, coins, fire and future samurai enemy. Keep the arena and UI in one palette: deep slate, warm orange, pale text, red hearts. Use restrained rectangular UI with pixel-art cues. Sound: Kenney Digital Audio (CC0), with separate shoot, hurt and pickup sounds. Retain licenses and source links; do not mix unrelated visual themes.

08 / MVP FOR THE WEEK-15 BUILD

One complete mobile loop: menu -> playable survival arena -> pause/resume/restart/menu. Responsive HUD with hearts, coin score, survival time and touch controls; notch-safe layout. Keep the existing spawning, health, cooldown, invulnerability and 60-second gate goal. Later milestones add the planned enemy AI, local best-score/settings saving and sound/effect polish. No networking or backend.

09 / CUT LIST

No multiplayer, accounts, online leaderboard, shop, ads, procedural levels, inventory, weapon upgrades or multiple worlds. No new 3D assets. Do not add more levels before the single arena works reliably on the three required device formats.

10 / PLAN: WEEKS 9-13 + ROLES

Week	Reviewable result
9	Kickoff: GDD, clean Git repository, menu/HUD/pause, safe-area layout and touch controls.
10	One patrolling enemy; verify attacks/collisions and balance the three falling types.
11	Save the local best score and sound preference; test quitting and reopening.
12	Polish three sound events, explosion feedback and clear damage feedback.
13	Feature freeze: fix device-layout issues, test Android, rehearse the English walkthrough.

Roles: solo project. The student owns design decisions, implementation, asset selection, testing and presentation. No team roles are assigned before week 14.

Kickoff evidence: HUD captures on a 16:9 phone, a notch phone and a 4:3 tablet; menu and pause captures; 3 meaningful commits and a merged feature PR. Submit a 2-3 minute English video with the student's own voice.